

Model

1 Model

1.1 Environment

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \dots, n\}$. These locations are partitioned into municipalities $g \in G$:

$$S = \bigcup_{g \in G} S^g, \quad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed supply of land $\bar{H}_i$, allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $\bar{H}_i^R$ is residential floor space capacity and $\bar{H}_i^F$ is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium.

Locations are connected by a transportation network. The generalized commuting cost from residence i to workplace j is

$$\tau_{ij} = \tau_{ij}(I_{ij}, n_{ij}),$$

where I_{ij} is transportation investment on link (i, j) and n_{ij} is the equilibrium commuter flow on that link. Investment reduces commuting costs and congestion raises them:

$$\frac{\partial \tau_{ij}}{\partial I_{ij}} \leq 0, \quad \frac{\partial \tau_{ij}}{\partial n_{ij}} \geq 0.$$

Following Fajgelbaum and Schaal, 2020, I parametrize the commuting cost as

$$\tau_{ij} = \delta_{ij}^\tau \frac{n_{ij}^\rho}{I_{ij}^\gamma},$$

where $\delta_{ij}^\tau > 0$ captures geographic features of link (i, j) and $\rho, \gamma > 0$.

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

1.2 Households

A household chooses a residence location i and a workplace j to maximize utility. Preferences over the numeraire good Q_{ij} and residential floor space h_{ij}^R are Cobb-Douglas:

$$u_{ij}(\omega) = \frac{B_i}{\tau_{ij}} \left(\frac{Q_{ij}}{\alpha} \right)^\alpha \left(\frac{h_{ij}^R}{1-\alpha} \right)^{1-\alpha} v_{ij}(\omega),$$

where $B_i > 0$ is a residential amenity at location i , $\alpha \in (0, 1)$ is the expenditure share on the composite good, and $v_{ij}(\omega)$ is an idiosyncratic preference shock drawn independently for each (i, j) pair.

The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R \leq w_j,$$

where q_i^R is the residential rent at location i , w_j is the wage at workplace j , and $\phi(N_i, L_i)$ is a monetary disutility from residing at i . The disutility captures two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial \phi}{\partial N_i} > 0, \quad \frac{\partial \phi}{\partial L_i} > 0,$$

where N_i denotes residential population and L_i denotes employment at location i (defined below). I parametrize the disutility as

$$\phi(N_i, L_i) = \eta_N N_i^{1+\rho} + \eta_L L_i^{1+\rho}, \quad (1)$$

where $\eta_N, \eta_L > 0$ are congestion weights on residential crowding and worker influx respectively, and $\rho > 0$ governs the convexity of congestion costs. The marginal congestion costs are $\partial \phi / \partial N_i = (1+\rho)\eta_N N_i^\rho$ and $\partial \phi / \partial L_i = (1+\rho)\eta_L L_i^\rho$, both increasing in their arguments. Convexity ($\rho > 0$) ensures the municipal first-order condition has an interior solution: as the municipality loosens residential zoning, each additional resident raises congestion by more than the last, so the congestion cost eventually dominates the rent benefit. The ratio η_N / η_L governs the relative weight incumbents place on residential crowding versus worker influx, and is disciplined by the observed mix of residential and commercial zoning restrictions across locations. The overall scale (η_N, η_L) is calibrated to match the housing supply elasticity estimates of Baum-Snow and Han (2024), which proxy for how tightly the municipality's zoning constraint binds at the equilibrium allocation.

Optimal demands are $Q_{ij}^* = \alpha Y_{ij}$ and $h_{ij}^{R*} = (1-\alpha)Y_{ij}/q_i^R$. Substituting into the utility function yields the deterministic component of indirect utility:

$$\tilde{V}_{ij} = \frac{B_i w_j}{\tau_{ij} (q_i^R)^{1-\alpha}}.$$

The idiosyncratic shock $v_{ij}(\omega)$ is drawn from a Fréchet distribution with $\Pr(v_{ij} \leq z) = \exp(-T_i E_j z^{-\epsilon})$, where $T_i > 0$ and $E_j > 0$ are location-specific scale parameters and $\epsilon > 1$ governs the dispersion of idiosyncratic tastes. By the Fréchet aggregation property, the probability that a household chooses residence i and workplace j is

$$\pi_{ij} = \frac{T_i E_j \tilde{V}_{ij}^\epsilon}{\Phi}, \quad \Phi \equiv \sum_{i'} \Phi_{i'}, \quad \Phi_i \equiv \sum_j T_i E_j \tilde{V}_{ij}^\epsilon, \quad (2)$$

and expected utility is

$$U = \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon}. \quad (3)$$

Let N denote total metropolitan population, fixed in a closed economy. Equilibrium residential population at each location is

$$N_i = N \sum_j \pi_{ij}.$$

Residential floor space market clearing at location i pins down the equilibrium rent:

$$q_i^R = \frac{(1 - \alpha) \sum_j N \pi_{ij} Y_{ij}}{\bar{H}_i^R}. \quad (4)$$

1.3 Firms

At each location j , competitive firms produce a freely tradable good using labor L_j and commercial floor space H_j^F :

$$Y_j = A_j L_j^\gamma (H_j^F)^{1-\gamma},$$

where $A_j > 0$ is location-specific productivity and $\gamma \in (0, 1)$. Taking wages w_j and commercial rents q_j^F as given, profit maximization yields:

$$w_j = \gamma A_j \left(\frac{H_j^F}{L_j} \right)^{1-\gamma}, \quad q_j^F = (1 - \gamma) A_j \left(\frac{L_j}{H_j^F} \right)^\gamma.$$

Commercial floor space is constrained by zoning: $H_j^F = \bar{H}_j^F$. Labor market clearing at j gives

$$L_j = N \sum_i \pi_{ij},$$

determined by household location choices. The disutility $\phi(N_i, L_i)$ uses L_i to denote employment at location i acting as a workplace, which is this same quantity evaluated at $j = i$.

1.4 Municipal Zoning

Each municipality $g \in G$ chooses the land-use allocation $\{\bar{H}_i^R, \bar{H}_i^F\}_{i \in S^g}$ to maximize the total welfare of its residents. In line with the political economy of local land use (Mast, 2024; Rollet and Weiwu, 2025), the municipal government acts as a representative of its residents, who face a tradeoff between lower housing costs — which favor denser zoning — and lower congestion and fewer workers near their homes — which favor restrictive zoning. Formally, municipality g solves

$$\max_{\{\bar{H}_i^R, \bar{H}_i^F\}_{i \in S^g}} W^g = \sum_{i \in S^g} N_i (\bar{U} - \phi(N_i, L_i))$$

where N_i is the equilibrium population at location i and $\bar{U}_i = \Gamma(1 - \frac{1}{\epsilon}) \Phi_i^{1/\epsilon}$ is the expected utility of a representative resident at location i , subject to $\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i$ for each $i \in$

S^g , taking other municipalities' allocations $\{\bar{H}_{i'}^R, \bar{H}_{i'}^F\}_{i' \notin S^g}$ and transportation investment $\{I_{jk}\}$ as given.

Substituting $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ and differentiating $\sum_{i \in S^g} N_i \bar{U}_i$ with respect to $\bar{H}_i^R$, the interior first-order condition at location $i \in S^g$ is

$$\underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\geq 0 \text{ (in-movers)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R}}_{\geq 0 \text{ (rent benefit)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial N_i} \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\leq 0 \text{ (congestion cost)}} + N_i \underbrace{\left(-\frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial L_i} \frac{\partial L_i}{\partial \bar{H}_i^R} \right)}_{\geq 0 \text{ (less worker influx)}} = 0. \quad (5)$$

The first term is positive: looser zoning lowers rents, attracting additional residents from the rest of the metro area; each in-mover contributes $\bar{U}_i$ to the municipality's objective. The second term is positive: lower rents raise effective income for all N_i existing residents. The third term is negative: in-movers raise N_i , worsening congestion for all current residents through ϕ . The fourth term is positive: shifting land from commercial to residential reduces employment L_i and the associated worker congestion. The municipality sets $\bar{H}_i^R$ where the benefit of attracting residents and lowering rents is offset by the congestion they impose.

1.5 Transportation Authority

The metropolitan transportation authority chooses investment $\{I_{jk}\}_{j,k \in S}$ to maximize aggregate household welfare, taking the full zoning vector $\{\bar{H}_i^R, \bar{H}_i^F\}$ as given:

$$\max_{\{I_{jk}\}} N \cdot \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon}$$

subject to the budget constraint

$$\sum_{j,k} \delta_{jk} I_{jk} \leq K, \quad I_{jk} \geq 0,$$

where δ_{jk} is the per-unit cost of investment on link (j, k) and K is the total infrastructure budget.

Let $\mu \geq 0$ be the Lagrange multiplier on the budget constraint. The interior first-order condition for link (j, k) is

$$\frac{N}{\epsilon} \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon-1} \frac{\partial \Phi}{\partial I_{jk}} = \mu \delta_{jk}. \quad (6)$$

The left-hand side is the marginal social benefit of investment on link (j, k) : lower commuting costs raise $\tilde{V}_{ij}$ for all commuters on that link, increasing Φ and hence aggregate welfare. The authority equates the marginal benefit per dollar of investment across all active links.

2 Empirical Facts

The model's first-order conditions generate two testable predictions. From the municipal zoning FOC equation (5): locations with better transit access face additional upward pressure on population, which amplifies the congestion term and triggers a preemptive tightening of zoning (**Proposition 1**). From the transportation authority's FOC equation (6): because the marginal social benefit of transit investment depends on ridership,

which depends on the density of residents and jobs near the corridor, restrictive zoning that suppresses station-area density should lower the equilibrium return to transit investment (**Proposition 2**).

I test both predictions using Dallas–Fort Worth (DFW) tract-level data. These tests are best understood as *equilibrium consistency checks* on the model’s mechanism, not as causal estimates of the effect of zoning on transportation outcomes; a separate identification strategy, using historical transit routes, planned-route instruments, or policy shocks to zoning authority, would be needed to establish causality. Both propositions find support within DFW, though neither can claim causal identification from the cross-sectional data, as I discuss below.

2.1 Data and Zoning Proxy

The primary proxy for zoning restrictiveness is the tract-level housing supply elasticity $\hat{\gamma}_i$ of Baum-Snow and Han (2024), available for the DFW metro area ($N \approx 590$ –1,016 tracts depending on the outcome). Lower $\hat{\gamma}_i$ indicates that the local housing stock responds less to demand shocks, corresponding to a tighter effective constraint on $\bar{H}_i^R$ in the model. Employment inflows L_i are constructed from LODES 2023 origin-destination flows, aggregated to the workplace tract. Transit access is measured using DART station locations (148 stations in DFW), with a tract classified as transit-adjacent if its centroid lies within 0.5 miles of a station (15 tracts) or 1 mile (86 tracts).

2.2 Proposition 1: Transit Access Triggers a Zoning-Tightening Response

The municipal FOC also implies that transit access, by raising residential attractiveness $\tilde{V}_{ij}$ and the resulting equilibrium population pressure N_i , should trigger a preemptive tightening of zoning: $\hat{\gamma}_i|_{\text{near transit}} < \hat{\gamma}_i|_{\text{far}}$. I estimate

$$\hat{\gamma}_i = \alpha + \beta_1 \mathbf{1}[\text{near DART}]_i + \beta_2 \text{pctdis}_i + \beta_3 \log(1 + L_i) + \varepsilon_i$$

on the subsample of $N = 581$ Baum-Snow tracts matched to 2020 Census geometry, of which 15 have a DART-adjacent centroid.

DART-adjacent tracts have housing supply elasticities roughly 0.15–0.21 units lower than non-adjacent tracts, significant at the 1–5% level across specifications (Table 1), and the result is robust to controlling separately for CBD distance and employment concentration. The interaction between transit proximity and CBD distance is not significant, likely reflecting the small number of suburban DART-adjacent tracts in the sample. This provides within-DFW support for the model’s congestion channel: transit access, which raises local population pressure in the model, predicts tighter observed zoning.

2.3 Proposition 2: Restrictive Zoning and the Station-Area Density Base

The transportation authority’s FOC implies that the marginal social benefit of transit investment depends on ridership, which in turn depends on the density of residents and jobs near the corridor. If zoning suppresses that station-area density, the model predicts a lower equilibrium return to transit investment. I test this mechanism directly at the tract level within DFW, using the interaction between the Baum-Snow and Han housing supply

Table 1: DART Access and Housing Supply Elasticity

	(1)	(2)	(3)	(4)
$\mathbf{1}[\text{near DART}]$	-0.152** (0.059)	-0.205*** (0.058)	-0.178*** (0.057)	-0.177** (0.079)
pctdis		-0.412*** (0.059)	-0.384*** (0.059)	-0.384*** (0.059)
$\log(1 + L_i)$			-0.032*** (0.007)	-0.032*** (0.007)
$\mathbf{1}[\text{near}] \times \text{pctdis}$				-0.011 (0.418)
R^2	0.011	0.087	0.118	0.118
N	581	581	581	581

Note: Standard errors in parentheses. Sample restricted to the 581 Baum-Snow tracts with matched 2020 Census geometry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

elasticity and DART proximity. The key prediction is not that more permissive zoning produces universally higher density—equilibrium sorting confounds this relationship in a cross-section—but that more permissive zoning should amplify the density that concentrates near transit nodes, since transit access and residential density are complements in the model’s agglomeration mechanism.

I estimate

$$\log(\text{density}_i) = \alpha + \beta_1 \hat{\gamma}_i + \beta_2 \mathbf{1}[\text{near DART}]_i + \beta_3 \hat{\gamma}_i \times \mathbf{1}[\text{near DART}]_i + X_i' \theta + \varepsilon_i \quad (7)$$

where $\hat{\gamma}_i$ is the Baum-Snow and Han housing supply elasticity, $\mathbf{1}[\text{near DART}]_i$ indicates a centroid within 0.5 miles of a DART station, and X_i includes CBD distance, log median income, and race shares. The model’s density channel implies $\beta_3 > 0$: near transit, additional permissiveness in zoning should raise density by more than elsewhere, because transit access magnifies the returns to residential concentration.

Columns (1)–(2) show that the direct elasticity-density gradient is negative and highly significant: tracts with lower supply elasticity (tighter zoning) are denser. This reflects an equilibrium confound rather than the model’s causal channel—central, high-demand locations in DFW are simultaneously denser and more politically restricted, and a static cross-section cannot separate these forces. Columns (3)–(5) show the interaction. The interaction coefficient $\hat{\beta}_3$ is positive and significant across all three specifications: 1.747** for population density at 0.5 miles, 1.507* for housing-unit density, and 1.487*** using the 1-mile buffer (Table 2). This means that near DART, the negative elasticity-density gradient is substantially attenuated: more permissive tracts achieve a notably higher density premium in the station area than more restrictive tracts. The implied elasticity-density slope is $-2.295 + 1.747 = -0.548$ near DART versus -2.295 far from DART (population density, column 3). The 1-mile buffer result, which covers 86 station-adjacent tracts and is therefore better powered, recovers a similarly sized interaction that is significant at the 0.1% level.

Taken at face value, the positive and significant $\hat{\beta}_3$ is consistent with the model’s density channel: restrictive zoning limits the station-area density concentration that would otherwise generate the ridership base for transit investment. The equilibrium confound in $\hat{\beta}_1$ underscores why this test cannot claim causal identification—a pre-determined or

Table 2: Zoning Permissiveness and Station-Area Density (Dallas, $N = 590$ tracts)

	(1)	(2)	(3)	(4)	(5)
Outcome:			log density		
	Pop., bivar.	Pop., ctrl.	Pop., 0.5 mi	Units, 0.5 mi	Pop., 1 mi
$\hat{\gamma}_i$	-1.709*** (0.153)	-2.189*** (0.132)	-2.295*** (0.133)	-2.549*** (0.132)	-2.384*** (0.138)
$\mathbf{1}[\text{near DART}]$			-0.923*** (0.231)	-0.841*** (0.229)	-0.531*** (0.139)
$\hat{\gamma}_i \times \mathbf{1}[\text{near DART}]$			1.747** (0.786)	1.507* (0.781)	1.487*** (0.368)
Controls	No	Yes	Yes	Yes	Yes
R^2	0.174	0.474	0.488	0.512	0.489
N	593	590	590	590	590

Note: Standard errors in parentheses. $\hat{\gamma}_i$ is the Baum-Snow and Han housing supply elasticity (higher = more permissive). Columns (1)–(2) show the baseline elasticity-density gradient. Columns (3)–(5) add the DART-proximity indicator and its interaction with $\hat{\gamma}_i$; column (4) uses housing-unit density as the outcome; column (5) defines “near DART” as a 1-mile buffer (86 DART-adjacent tracts vs. 15 in the 0.5-mile definition). Controls include CBD distance, log median household income, and minority-group population shares. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

exogenous zoning measure, or a panel design exploiting the timing of transit openings, would be needed to establish the direction of causality. These results nonetheless provide within-DFW equilibrium support for the mechanism: the degree to which transit proximity is associated with elevated density is lower in more restrictively zoned tracts, consistent with the transportation authority’s FOC.

2.4 Summary

Proposition 1, from the municipal zoning FOC, is strongly supported within DFW: DART-adjacent tracts are significantly more restrictively zoned, consistent with the model’s transit-pressure channel. Proposition 2, from the transportation authority’s FOC, is tested at the tract level using the interaction between Baum-Snow and Han housing supply elasticity and DART proximity. The direct elasticity-density relationship is confounded by equilibrium sorting, but the interaction term is positive and significant across all buffer definitions and outcome measures: near DART, more permissive tracts achieve a substantially larger density concentration than more restrictive tracts, consistent with the model’s prediction that restrictive zoning limits the station-area density base that determines transit investment returns. A credible causal test would require a pre-determined or exogenous zoning measure, or a panel exploiting the timing of transit openings; the present cross-sectional results provide equilibrium discipline for the mechanism rather than a causal estimate.

References

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